9B Lesson 4

Plant products: starch, lipids and proteins

Year 9 Science · 9B Plant growth · Student Book pages 26 to 27

Name _____ Class _____ Date _____

Today I can:

- ☐ I can describe how plants use glucose to make starch, cellulose, lipids and proteins, and explain why nitrates are needed.
- ☐ I can test a leaf for starch and explain what happens when a seed germinates.

Key words 关键词

starch	淀粉
polymer	聚合物
cellulose	纤维素
lipid	脂质
protein	蛋白质
amino acid	氨基酸
enzyme	酶
nitrate	硝酸盐
germination	发芽
iodine solution	碘液
Benedict's solution	本尼迪克特试剂

Command words 指令词

Complete 完成（填写） · Describe 描述 · Draw 画出 · Explain 解释 · Label 标注 · Match 连线（配对） · Name 说出名称 · Predict 预测 · Use 使用（运用）

● Core Everyone does this section.

1. Practical: testing a variegated leaf for starch.

a) Draw the leaf before the test and after adding iodine. Label the green and white parts on the first drawing, and the blue-black parts on the second.

Before the test	After adding iodine

b) Name the solution used to test for starch and the colour it turns when starch is present.

c) Explain why only some parts of the leaf changed colour.

2. Complete the sentences. Use the word bank.

Glucose molecules made in photosynthesis are joined together to make a polymer called _____.

Starch is an energy store. It is found in the chloroplasts, in storage organs such as _____, and in seeds.

Sugars are also used to make _____, which builds plant cell walls, and _____ such as the oils in seeds.

3. Match each substance to one of its uses in a plant. Draw a line.

starch	speed up reactions such as photosynthesis and respiration
cellulose	energy store in seeds and potatoes
lipids	build cell walls
proteins	built from amino acids using nitrogen from nitrates
enzymes	make the waterproof cuticle and cell membranes

●● Stretch When Core is finished.

S1. Seeds contain a store of starch. Describe what happens when a seed germinates. Use the words water, enzymes, glucose, respiration and warm.

S2. Describe a test that would show that glucose is produced in a germinating seed. Include the colour change and one safety precaution.

S3. Explain why plants do not grow well when there is a lack of nitrates in the soil.

●●● Challenge Explain in full sentences.

C1. A plant was left in the dark for 24 hours. A metal foil stencil was then attached to one leaf and the plant was left in bright light for a day. The leaf was then tested for starch.

a) Explain why the plant was left in the dark first.

b) Predict the result of the starch test and explain it.

c) Explain why a metal foil stencil was used and not a clear plastic one.

C2. Avocado fruit flesh is rich in lipids. Avocado seeds also contain lipids. Explain why the plant puts lipids in each place. The two reasons are different.

My thinking (any language 可用中文)

A potato is a store of starch made by the plant in summer. When does the potato plant use this store, and why does it store starch rather than glucose?

Word bank starch 淀粉 · polymer 聚合物 · cellulose 纤维素 · lipids · potatoes · protein 蛋白质 · amino acid 氨基酸 · enzyme 酶 · nitrate 硝酸盐 · germination 发芽 · iodine solution 碘液

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Exit ticket 9B Lesson 4: Plant products: starch, lipids and proteins Name _____

1. Which solution tests for starch, and what colour does it turn?

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2. Why do seeds need a store of starch?

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3. What are enzymes made of, and which mineral salt do plants need to make them?

Today I can: Goal 1 ☐ Goal 2 ☐